

UNIVERSITAS
GADJAH MADA

Solution Strategies

Utilization, Nullification, Elimination

Problem Solving Methods and Modeling

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Unified Structured Inventive Thinking (USIT)

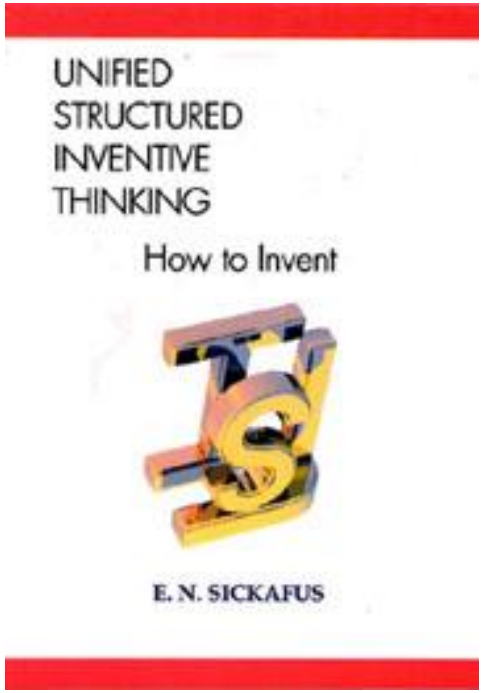


What is USIT

Unified Structured Inventive Thinking is a problem-solving method developed as a simplification of TRIZ.

Solution Strategies

In USIT, Sickafus also developed a problem-solving strategy based on **Utilization, Nullification, and Elimination**.



Utilization

Utilizing existing resources within the system to solve a problem without adding new elements.

Nullification

Neutralizing unwanted effects by reversing or eliminating their impact.

Elimination

Removing parts of the system that cause the problem or finding a solution that eliminates the need for certain elements

Utilization

The approach of utilizing existing resources within a system to solve problems without adding new elements. Aligns with the Closed-World Method principle, where solutions must be found using the minimal set of objects within the system.

In Closed-World Analysis, existing objects are analyzed to identify attributes and functions that can be utilized to improve the system without adding new components



Sample Case

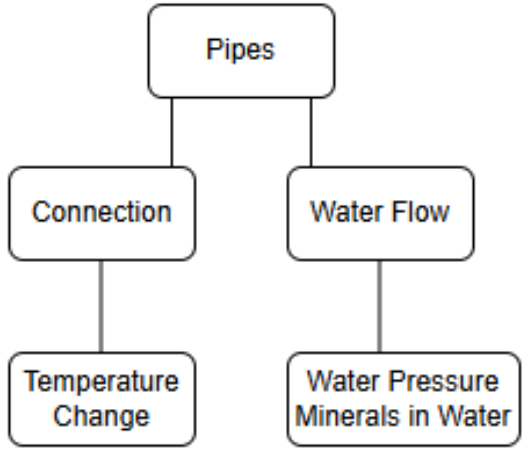
Reducing Water Leaks in Pipes Without Adding New Materials

With the Closed-World Method and Utilization approach, we seek solutions without introducing new materials from outside the system.

Case Resolution

1. **Minimal Set of Objects**
   - Pipe
   - Joint
   - Water Flow
   - Pressure in Pipe
2. **Utilization Approach**
   - Utilizing the water pressure itself
   - Taking advantage of the pipe's temperature changes
   - Harnessing the minerals in the water

3. **Closed-World Diagram**



Using water pressure to seal leaks

By adjusting the pressure within the pipe, small leaks can be sealed by the back pressure of the water itself.

Utilizing pipe temperature changes

If the pipe's temperature increases, the material around the joints will expand, naturally sealing the leak.

Harnessing minerals in the water

If the water contains lime or other minerals, these minerals can accumulate at the leak point and naturally seal it over time.

Exercise Utilization



Railway track bearings experience wear due to high friction between train wheels and the rails. Typically, to reduce friction and wear, external lubricants such as grease or oil are used



Find a solution without adding external lubricants, but by utilizing the existing materials within the system.

Nullification

Neutralizing the negative effects of a problem without eliminating its source. This closely resembles the concept of Transduction, where changes are made by redirecting problematic attributes or functions to an alternative pathway.

Case Example:

Overcoming Caffeine Side Effects Without Stopping Coffee Consumption



With the **Transduction technique** and **Nullification solution**, coffee consumption will not be stopped, but its negative effects will be neutralized, allowing the benefits of caffeine to be enjoyed without disruption.

Case Resolution

1. **Problem:** Caffeine is still consumed
2. **Negative effects:** Side effects such as heart palpitations and increased stomach acid.
3. **Nullification Approach**

   Not stopping coffee consumption, but neutralizing its negative effects to continue enjoying the benefits of caffeine without disruption.



1. Consume magnesium-rich sources (e.g., bananas, avocados, or magnesium supplements) to neutralize excessive stimulation on the heart.
2. Drink coffee after meals to reduce its impact on the stomach or Drink a glass of water after coffee to replenish lost fluids.

Exercise Nullification



1. **When an aircraft lands, its wheels experience a significant impact force, causing strong shocks that can:**
   - Endanger passengers due to severe jolts.
   - Cause premature wear on the wheels and aircraft frame.
   - Increase maintenance costs due to high stress on the suspension system



Use the Nullification approach to Neutralizing or reducing the negative effects of a problem without eliminating its source.

Elimination

Eliminating the source of the problem by removing elements that cause undesirable effects. This aligns with the concepts of Pluralization and the Closed-World Method, which emphasize simplifying the system by eliminating unnecessary components.

Sample Case

Eliminating Physical Textbooks with E-Books and Learning Management Systems (LMS)



The conventional solution is usually to reprint books in a lighter or more affordable edition, but this still does not address the root of the problem

Case Resolution



1. **Main Cause:** The use of physical textbooks, which are inflexible and expensive.
2. **Negative Impact:** Increased student burden, high costs, and limited access.
3. **Elimination Approach:** Instead of making textbooks lighter or cheaper, we completely remove physical books and replace them with e-books and a Learning Management System (LMS).

1. Eliminating the need for physical textbooks by replacing them with e-books (Easy Access).
2. Utilizing a Learning Management System (LMS) to store and manage learning materials online.
3. Enabling content updates without the need to reprint books
4. Easily accessible from any device (laptop, tablet, or smartphone).

Exercise Elimination



A company manages its own physical servers to run applications and store data. However, they face several issues:

- High operational costs
- Lack of flexibility
- High downtime risk



Use the **Elimination** approach to eliminate the negative impact of shocks without adding new materials or components.

CONCLUSION

These three strategies offer a more efficient approach to problem-solving compared to conventional solutions. The choice of strategy depends on the nature of the problem:

If existing resources can be utilized → **Use Utilization**

If negative effects can be eliminated without changing the cause → **Use Nullification**

If the root cause can be completely removed → **Use Elimination**

Any Questions?